Oral Cavity



ORAL CAVITY

Boundaries:

Roof – Floor – Sides – Ant & Post

Divisions:

1. Cavity proper 2. Vestibule

Contents

1. Tongue 2. Teeth

Ducts of Glands:

Nerve Supply:

Arterial Supply:

Lymph Drainage:



Boundaries

Roof: palate

1. Hard palate: anterior 2/3
2. Soft palate: posterior 1/3

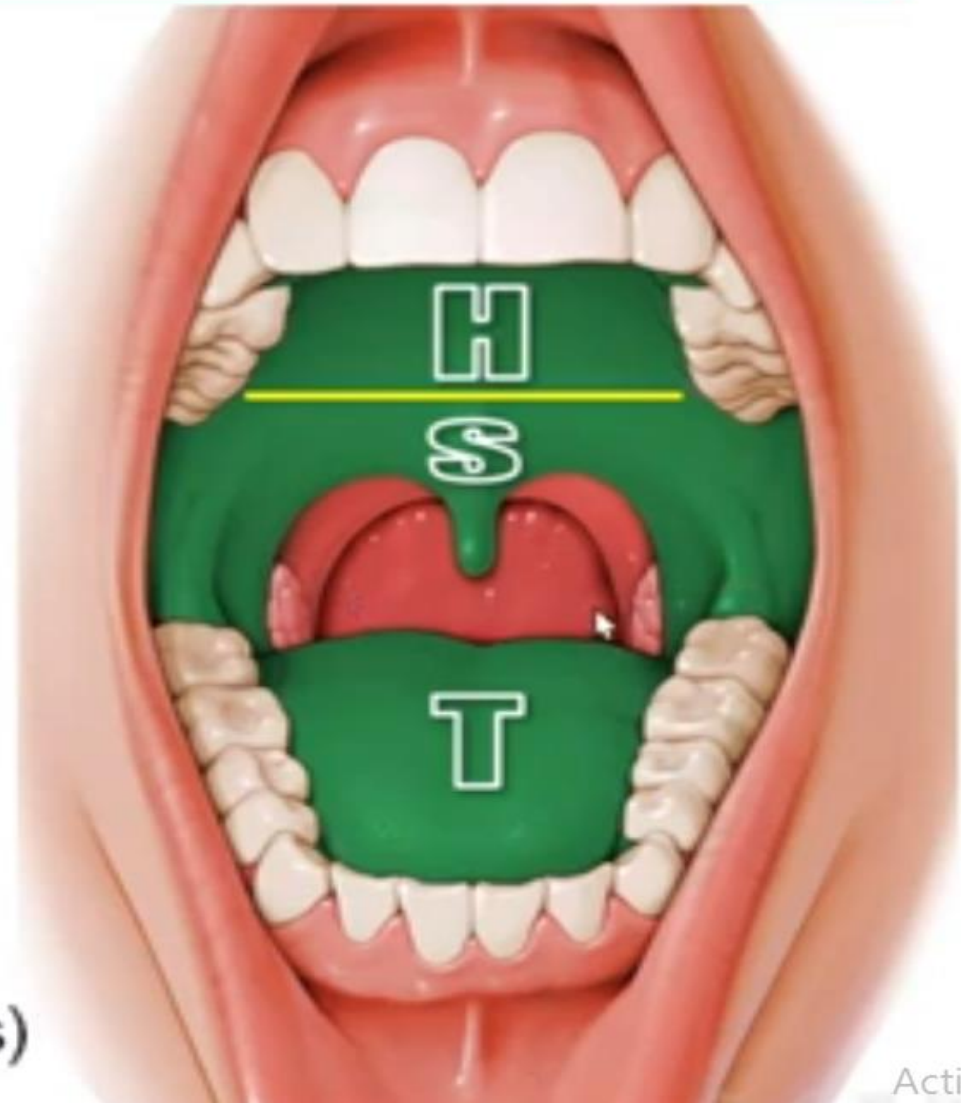
Floor:

1. anterior 2/3 of the tongue
2. Floor of the mouth

Front and Sides:

1. Cheeks
2. Lips

Posterior: Oropharynx (isthmus)



Acti

Divisions of the Oral Cavity

Mouth Cavity Proper

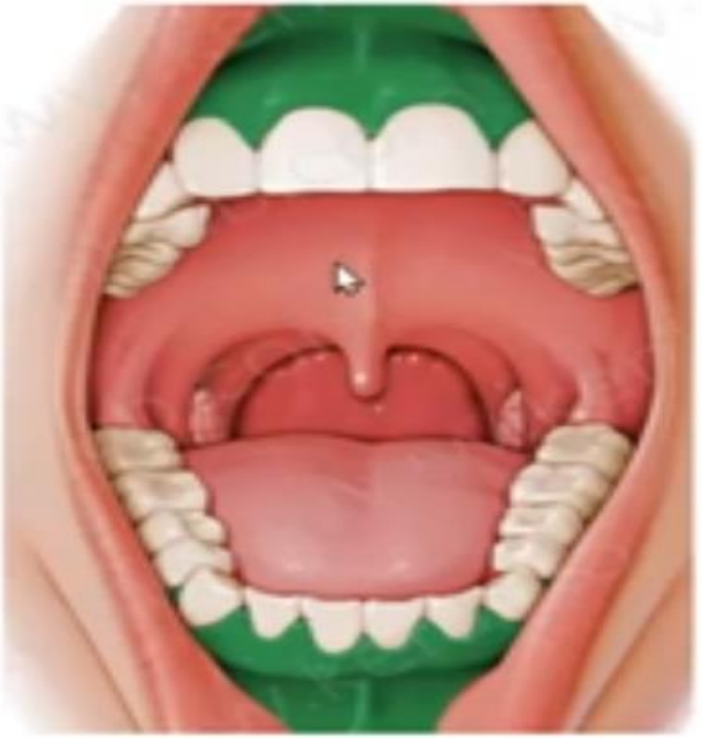
Lies inside the alveolar arches (teeth and gums)

Vestibule

Lies outside the alveolar arches (teeth and gums)







Activ
Go to

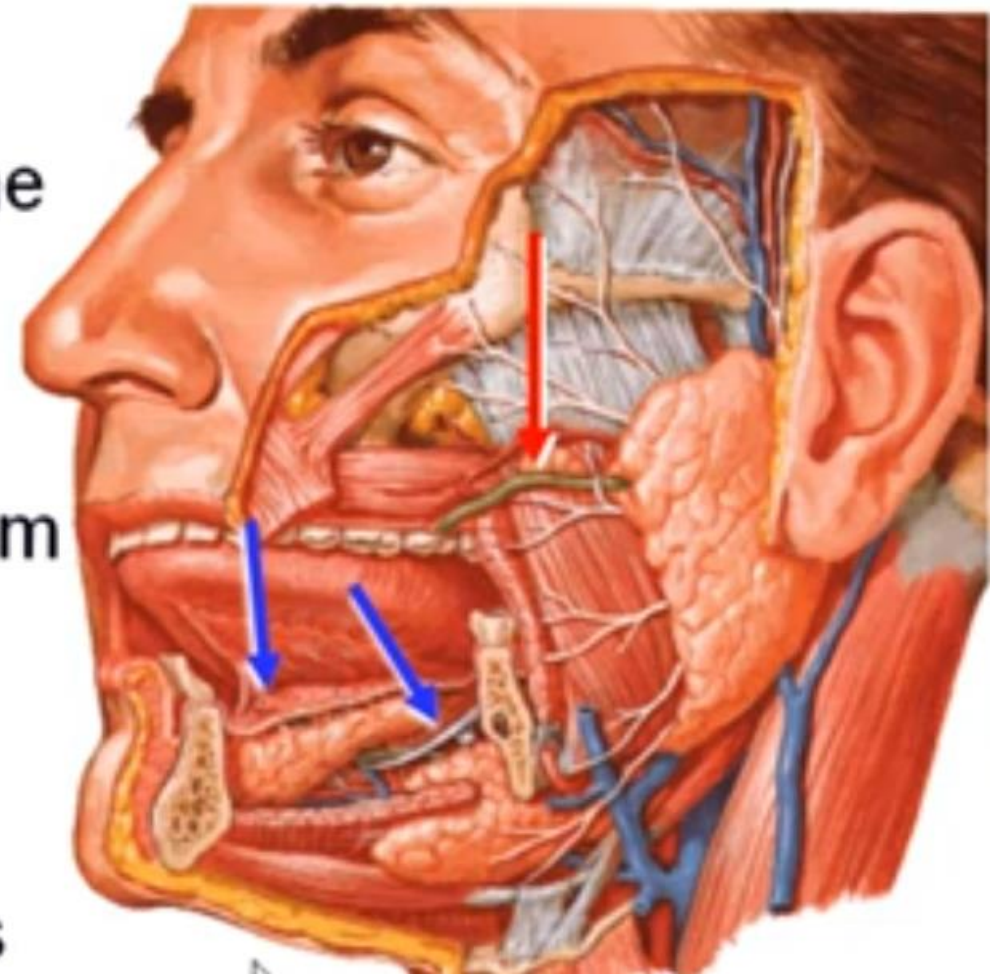
Ducts Opening Into the Oral Cavity

1. Parotid duct:
in the vestibule opposite the upper 2nd molar tooth

2. Submandibular duct:
on each side of the frenulum

3. Sublingual ducts: 10 - 12
on the sublingual fold

4. Mucous glands: numerous

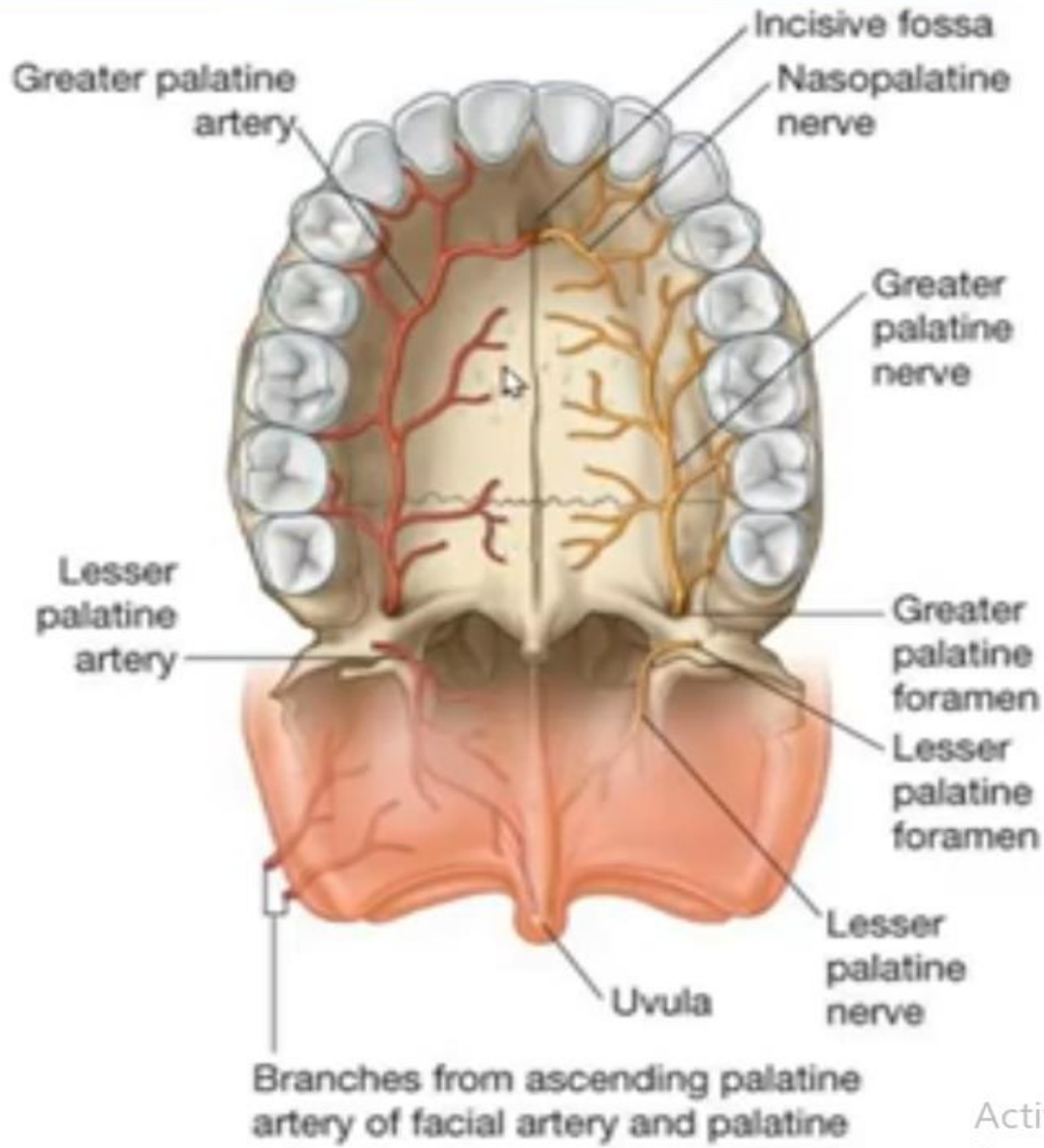


Nerve Supply of the Oral Cavity

1. Roof: Palate

2. Floor:

3. Cheek:



Nerve Supply of the Oral Cavity

1. Roof: Palate

sphenopalatine ganglion of the **maxillary nerve**

A. Greater palatine nerve

B. Lesser paltine nerve

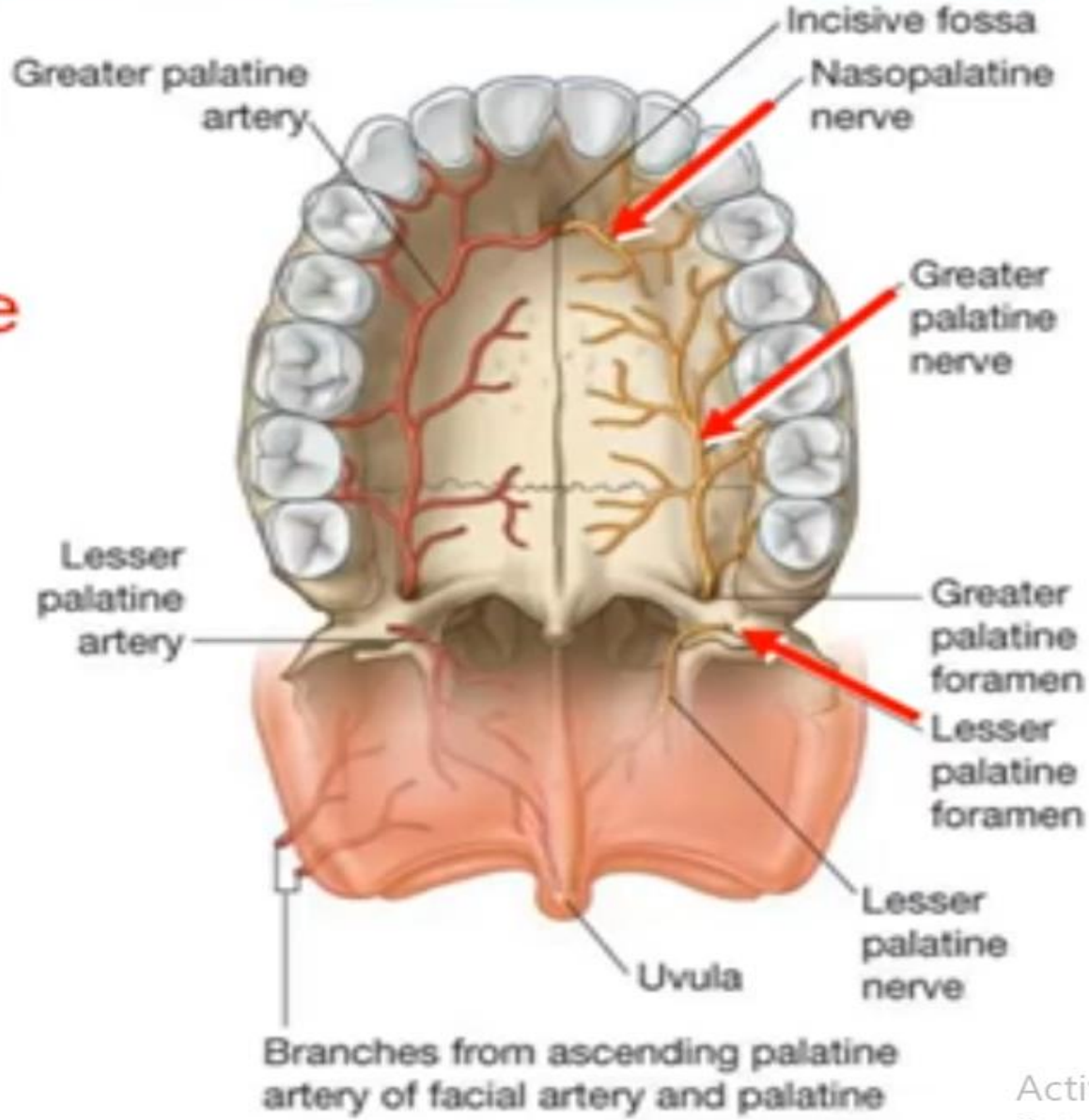
C. Nasopalatine nerve

2. Floor:

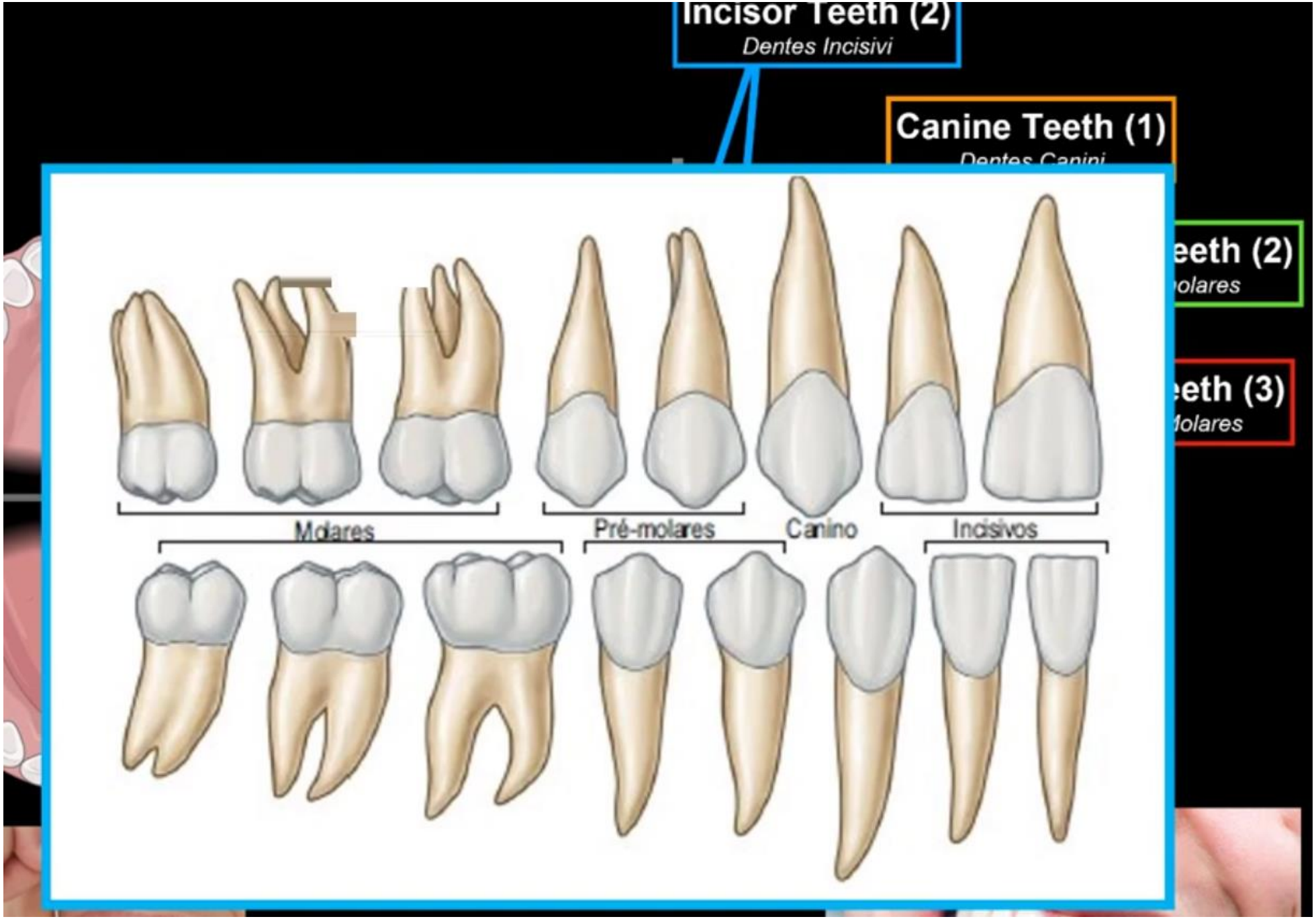
Lingual nerve from the mandibular nerve

3. Cheek:

Buccal nerve from the mandibular nerve



07:10


Incisor Teeth (2)
Dentes Incisivi
Canine Teeth (1)
eeth (2)
olares
eeth (3)
olares
Molares
Pré-molares
Canino
Incisivos

Nerve Supply of the Teeth

Maxillary Teeth:

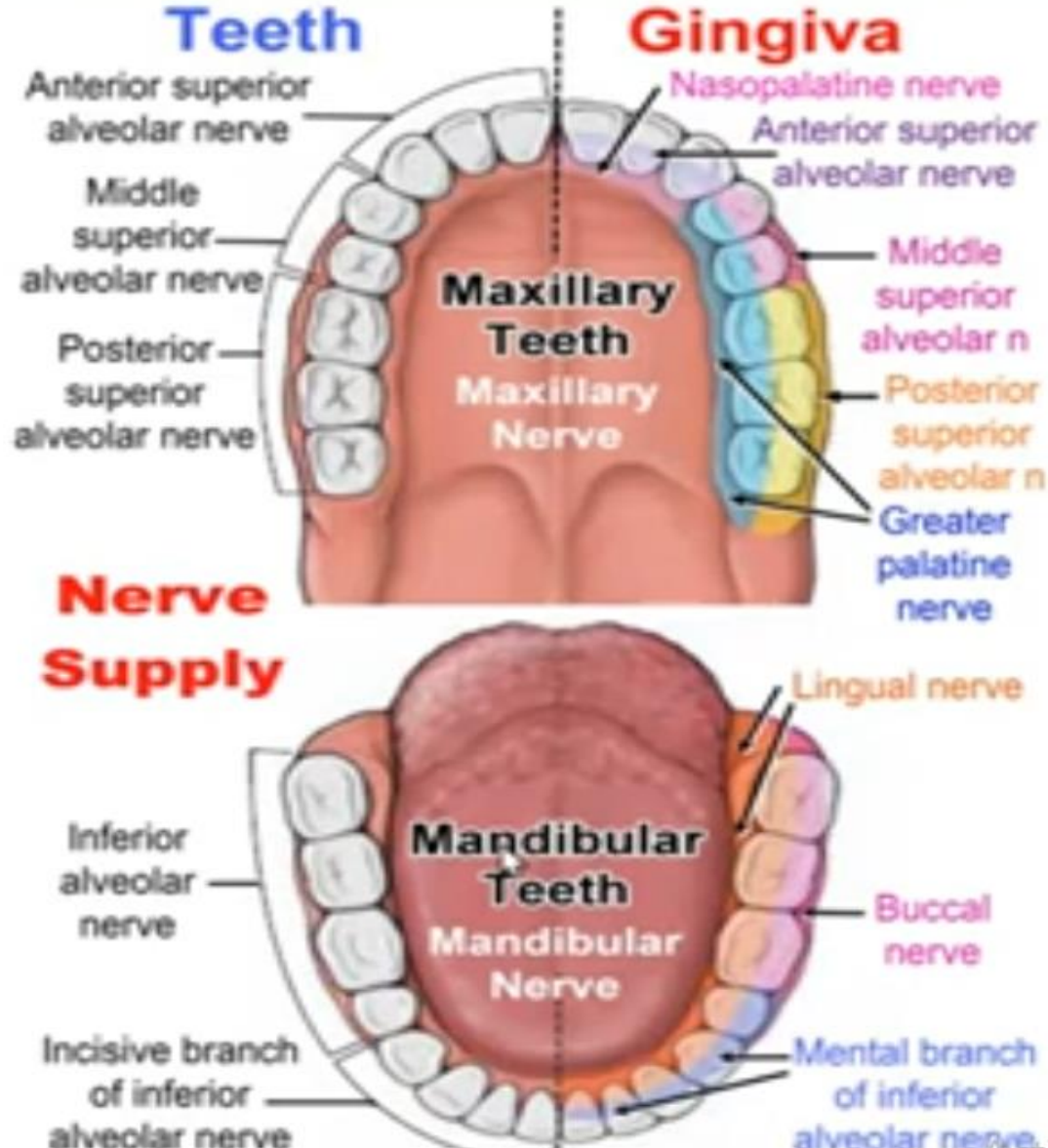
By the maxillary nerve

1. Ant sup alveolar nerve:
   the incisors and canine
2. Middle sup alveolar nerve:
   the premolar teeth
3. Post sup alveolar nerve:
   the molar teeth

Mandibular Teeth:

By the mandibular nerve

Inferior alveolar nerve and its incisive branch



Nerve Supply of the Gingiva

Maxillary Gingiva:

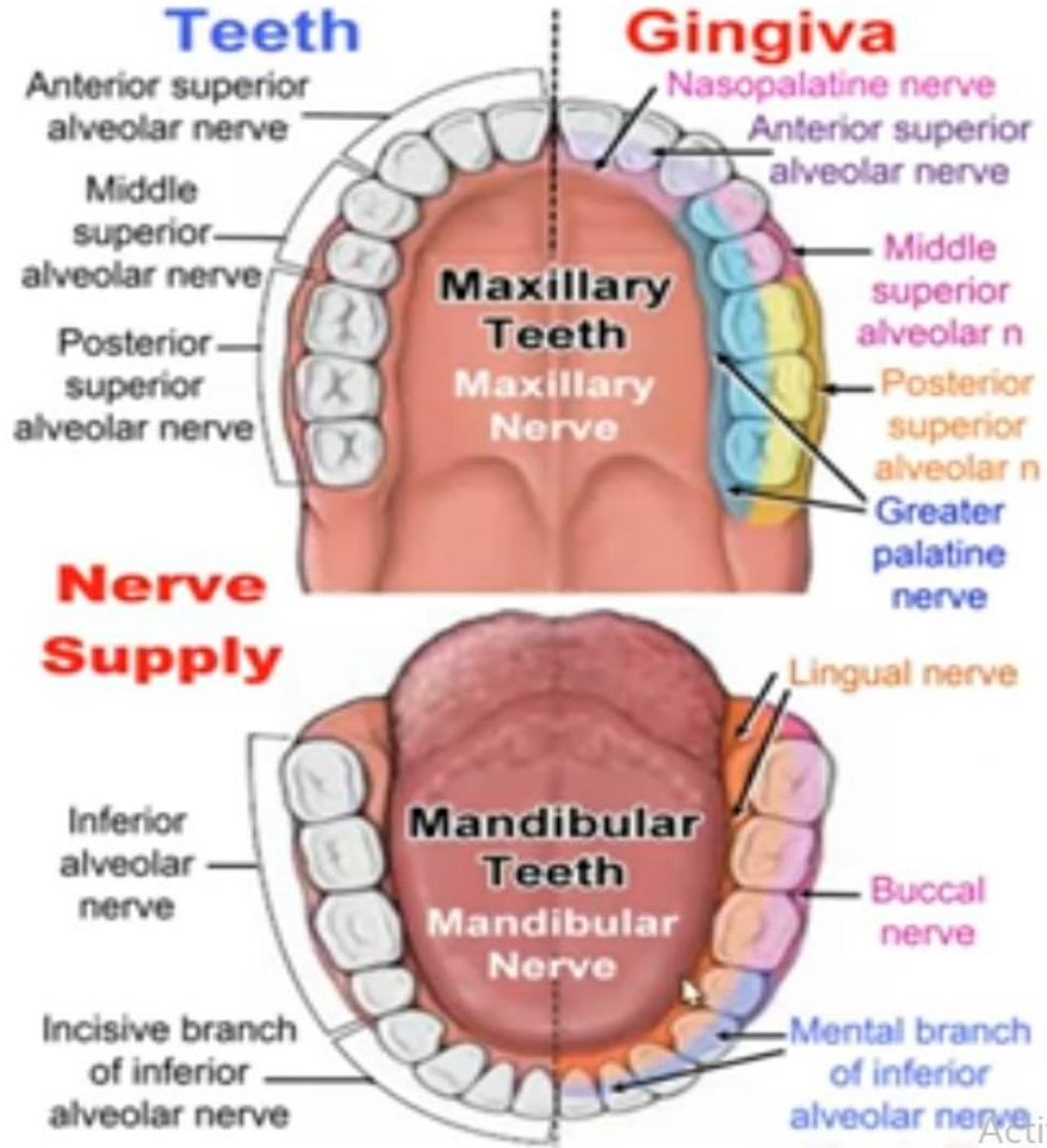
By the maxillary nerve

1. **Buccal Surface:** by the anterior, middle & posterior alveolar nerves
2. **Lingual Surface:** by the nasopalatine and greater palatine nerves

Mandibular Gingiva:

By the mandibular nerve

1. **Buccal Surface:** by the buccal and mental nerves
2. **Lingual Surface:** by the lingual nerve



08:02

ARTERIAL SUPPLY OF THE ORAL CAVITY

Hard Palate: maxillary artery

1. Greater palatine artery
2. Sphenopalatine artery

Soft Palate:

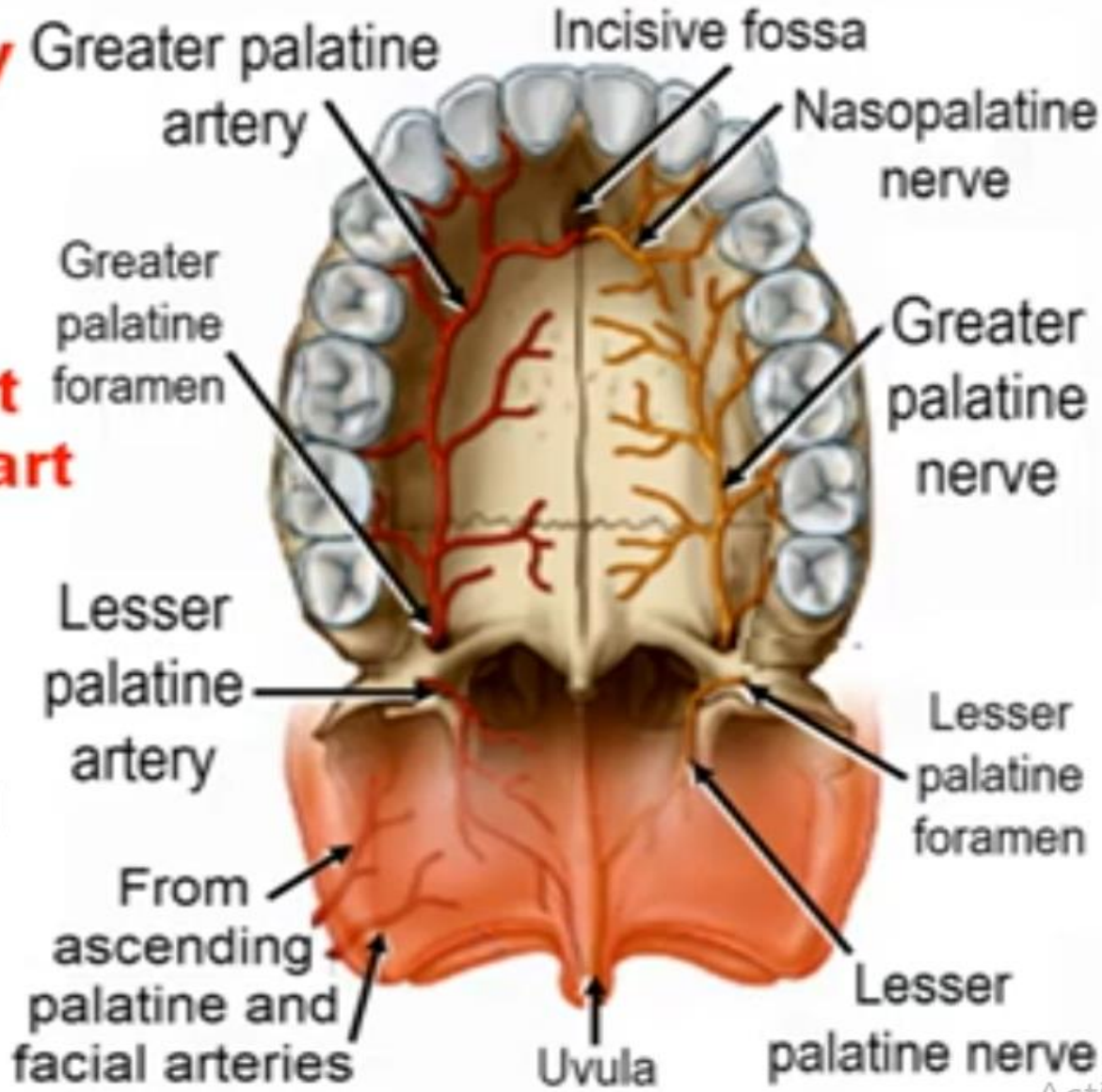
1. Lesser palatine art: **max art**
2. Ascending palatine: **facial art**
3. Palatine branches of the ascending pharyngeal art

Lips and Cheek: facial artery

Superior labial & inferior labial

Floor: lingual artery

1. Dorsal lingual arteries
2. Sublingual artery



THE TONGUE

Structure:

1. **Mucous membrane**
2. **Muscles**

Divisions: 2 parts

1. **Oral part: body**
   anterior 2/3
2. **Pharyngeal part: root**
   posterior 1/3

Muscles: 8 muscles

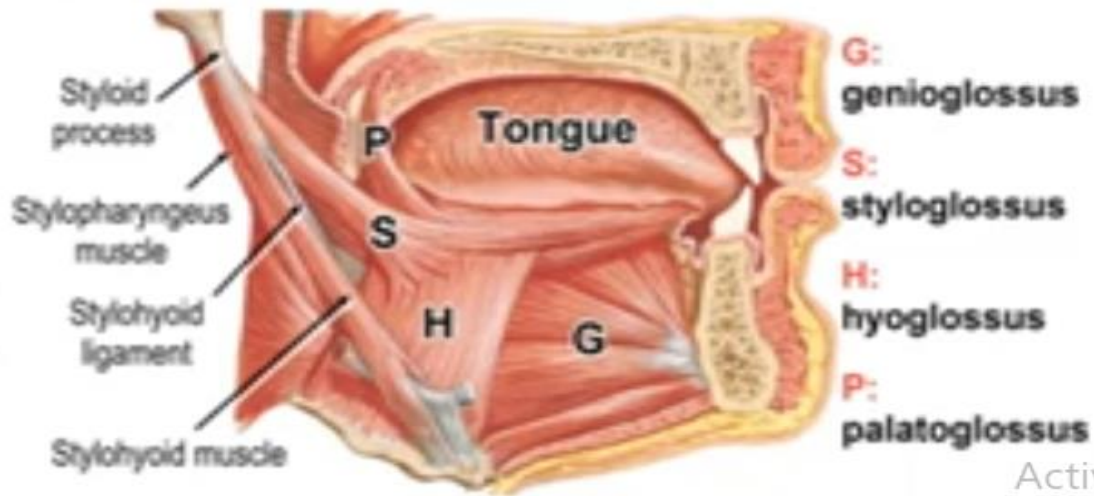
1. **Extrinsic muscles: 4**
2. **Intrinsic muscles: 4**

Blood Supply: lingual art & vein

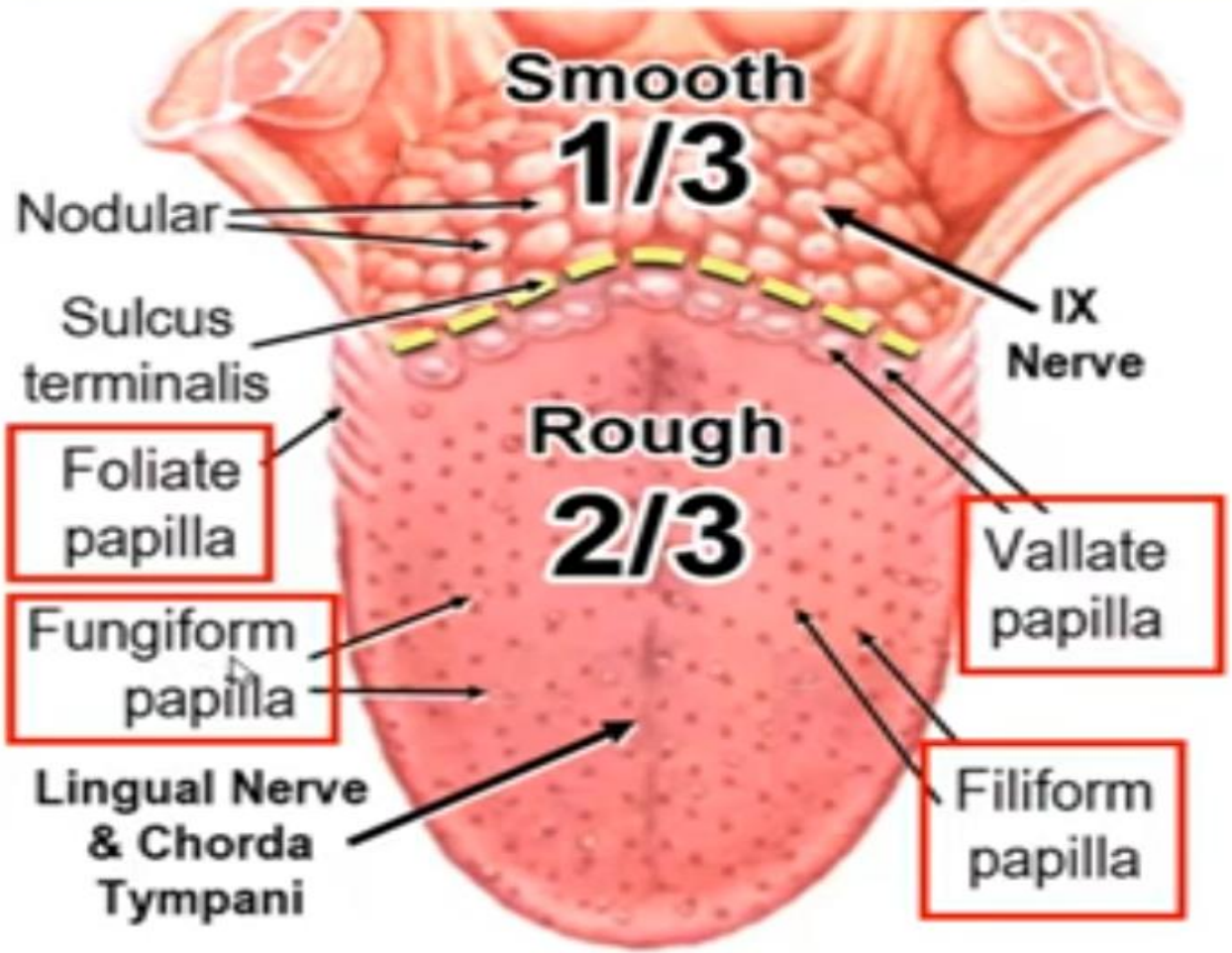
Nerve Supply: motor & sensory

Lymphatic Drainage:

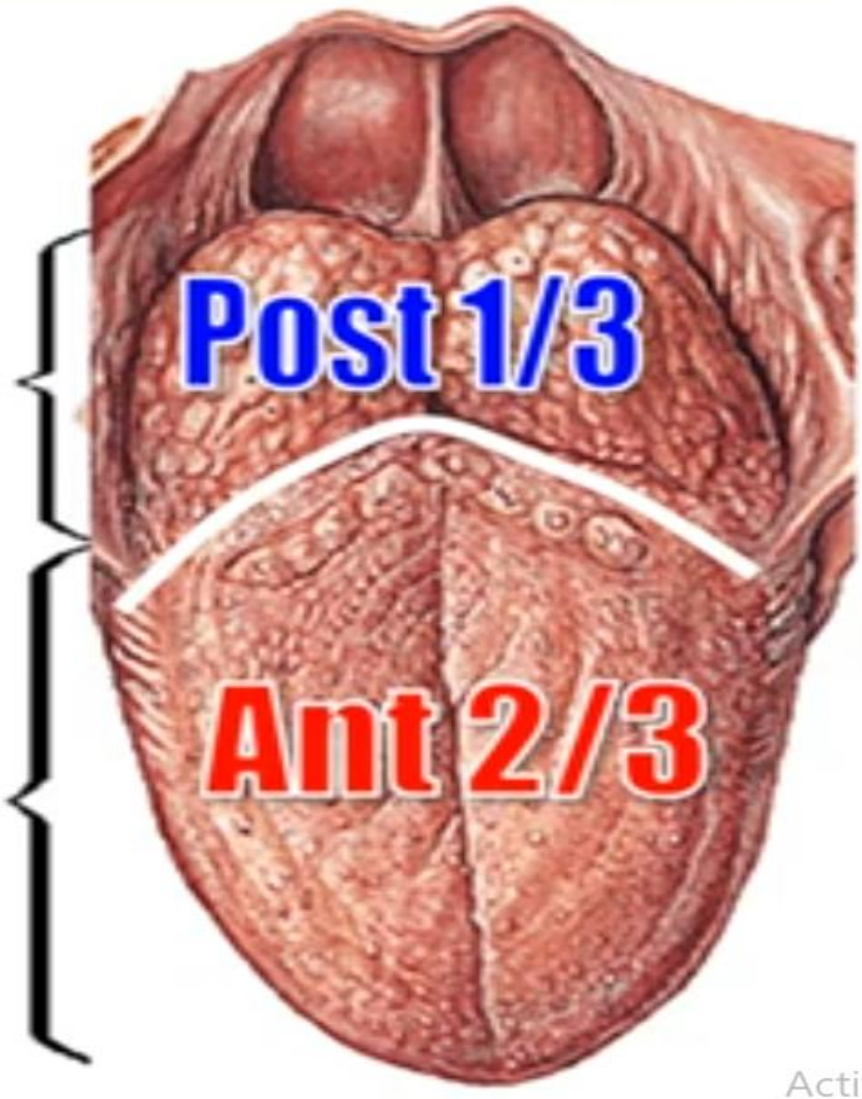




MUCOUS MEMBRANE



DORSAL SURFACE



Compare Between Oral & Pharyngeal Part

	ORAL	PHARYNGEAL
SIZE	Anterior 2/3	Posterior 1/3
POSITION	In the oral cavity	In the pharynx
Mucous Membrane	**Rough** due to presence of papillae	**Smooth** due to absence of papillae
Lymphoid Follicles	Absent	**Present** & the surface is nodular. Called **lingual tonsil**
Nerve Supply	Lingual & chorda tympani	Glossopharyngeal nerve
Development	1st arch	3rd arch
Arterial Supply	Deep lingual artery	Dorsal lingual arteries

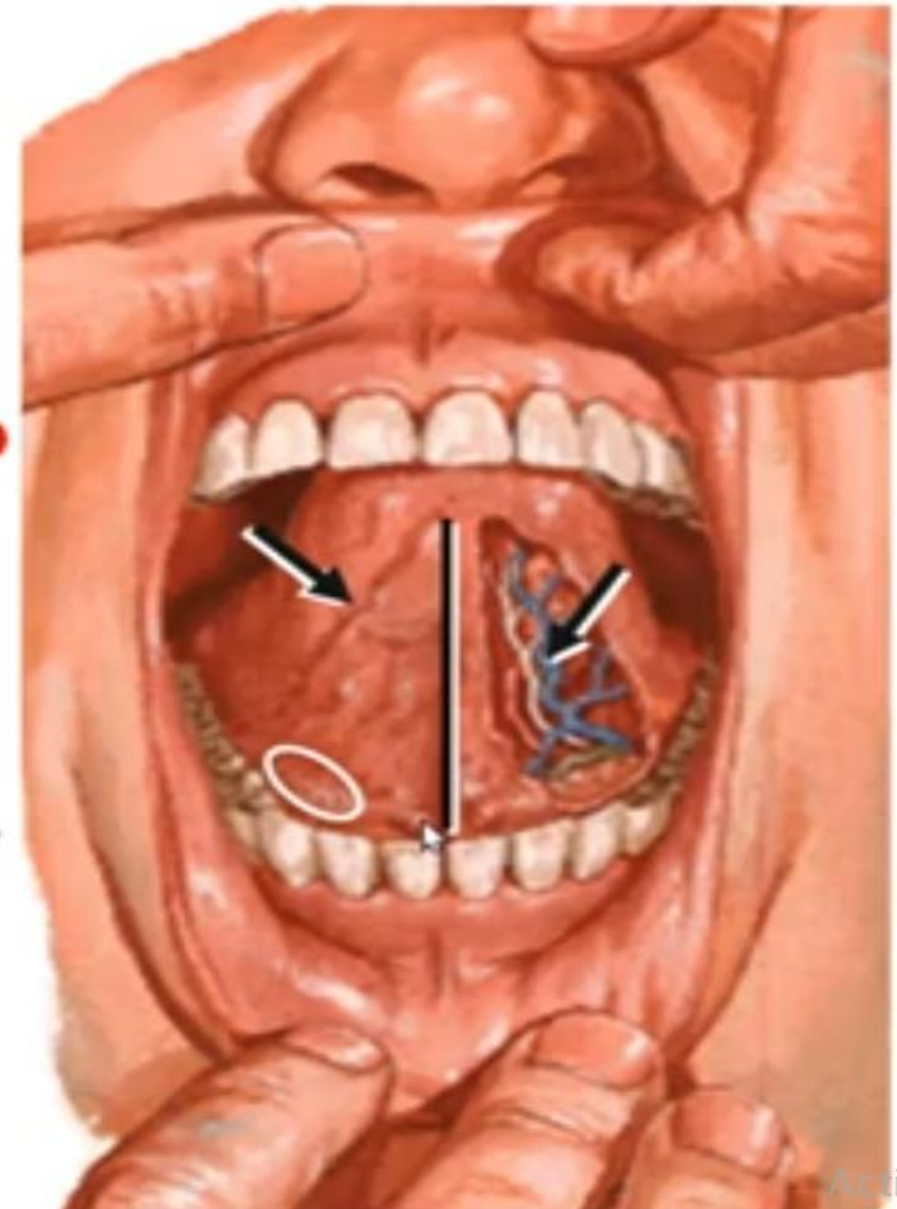
Ventral Surface of the Tongue

In the Midline:

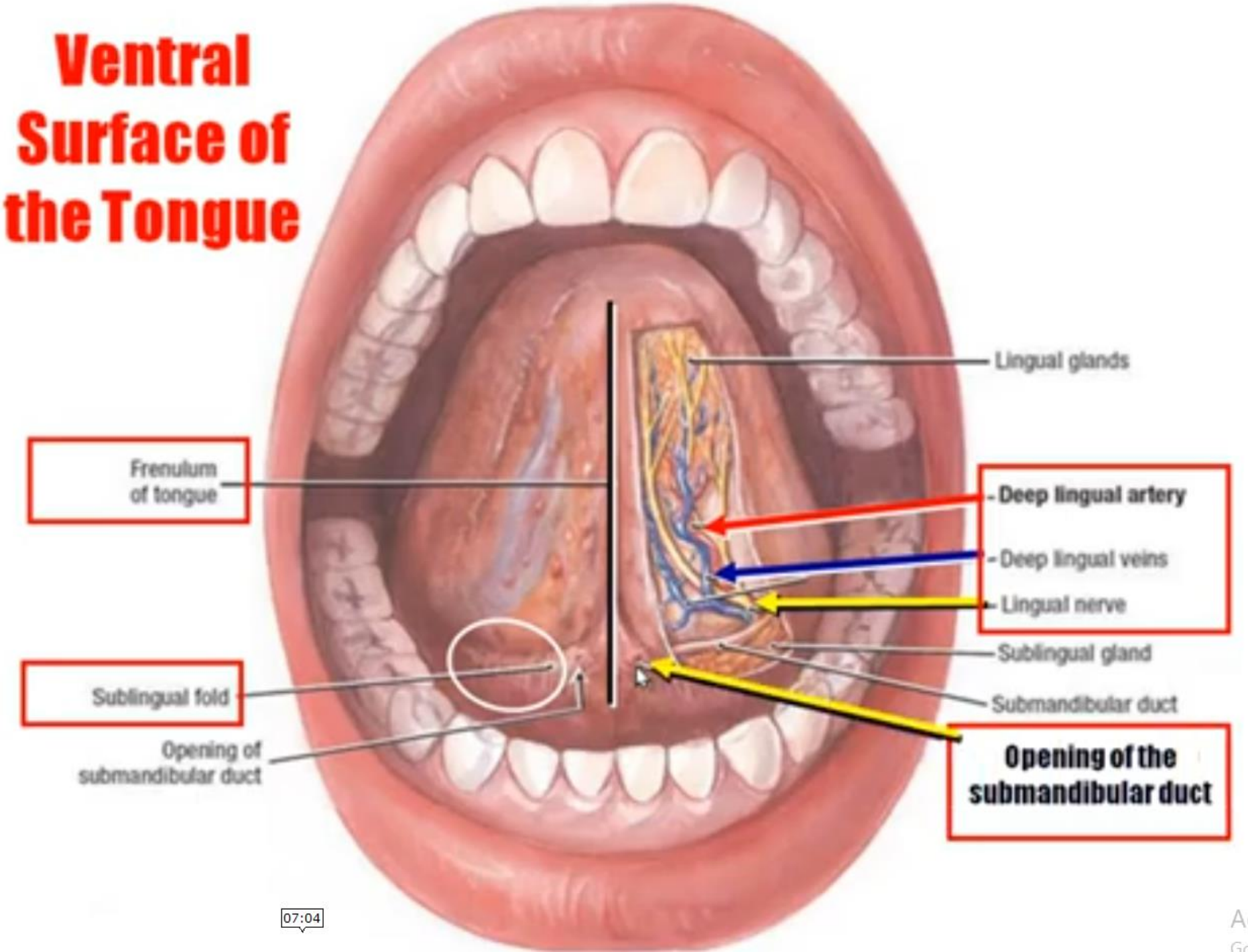
Frenulum of the tongue

On Each Side:

1. **Deep Lingual Vein,** with the **deep lingual artery** & **lingual nerve**
2. **Fimbriated fold**
3. **Sublingual fold** and the openings of the **sublingual ducts**
4. **Opening of the submandibular duct**

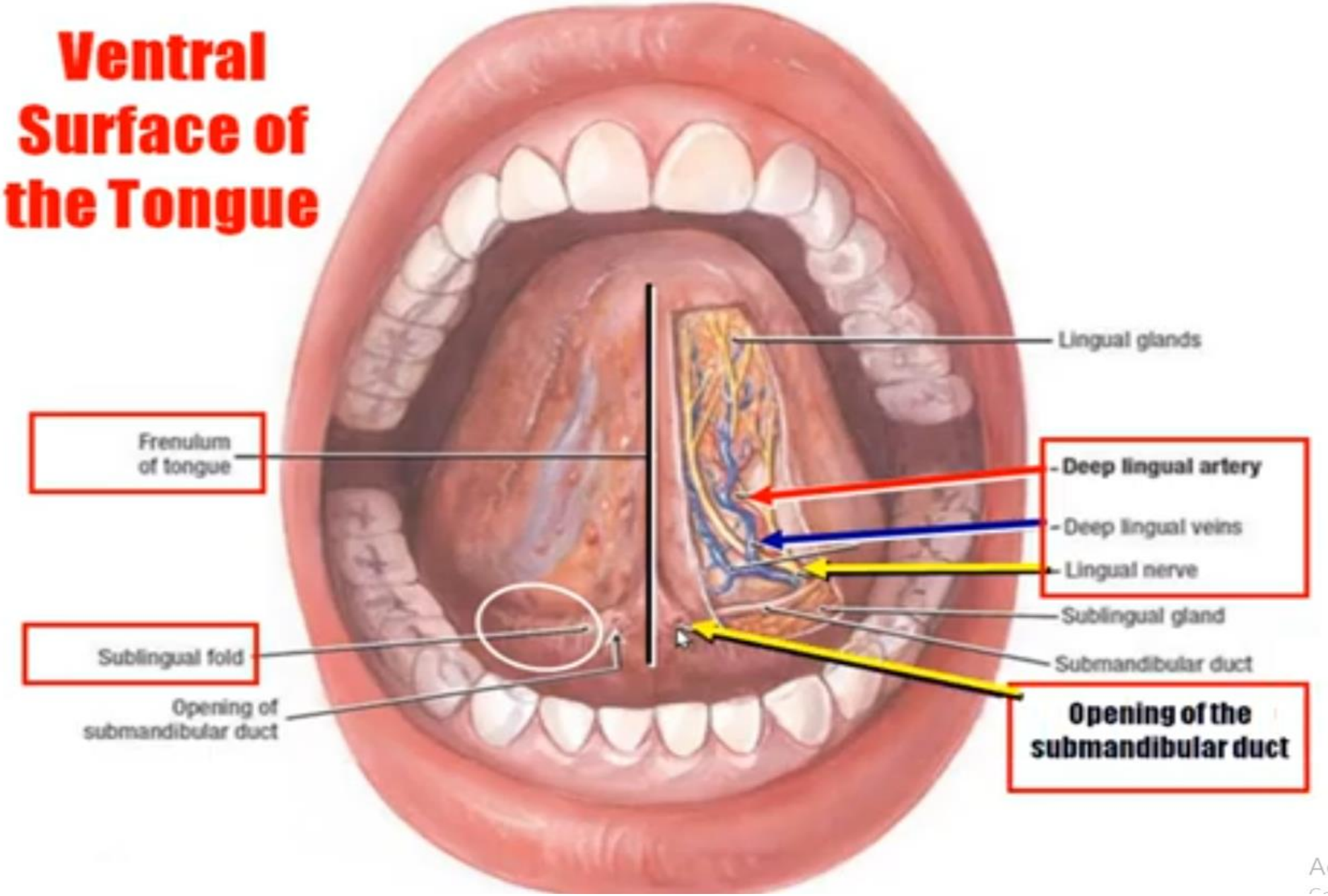


Ventral Surface of the Tongue



07:04

Ventral Surface of the Tongue

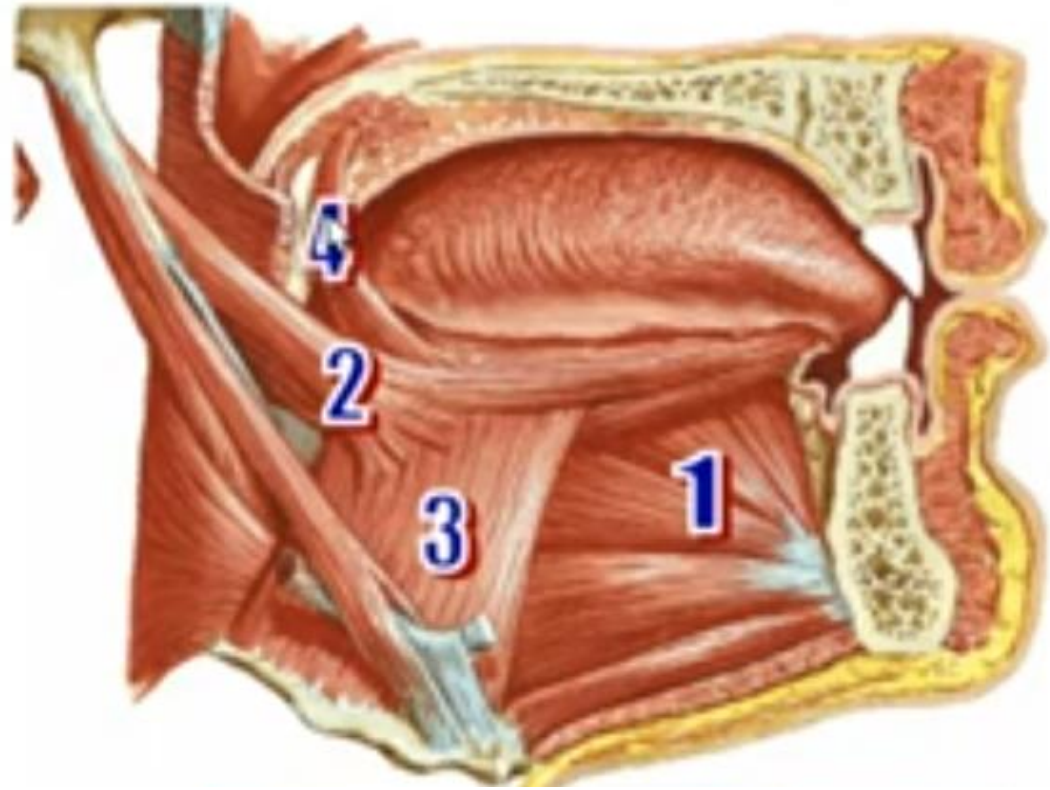


8 MUSCLES

4 Extrinsic & 4 Intrinsic

Extrinsic Muscles

1. Genioglossus: protrusion
2. Styloglossus: retraction
3. Hyoglossus: depression
4. Palatoglossus: elevation



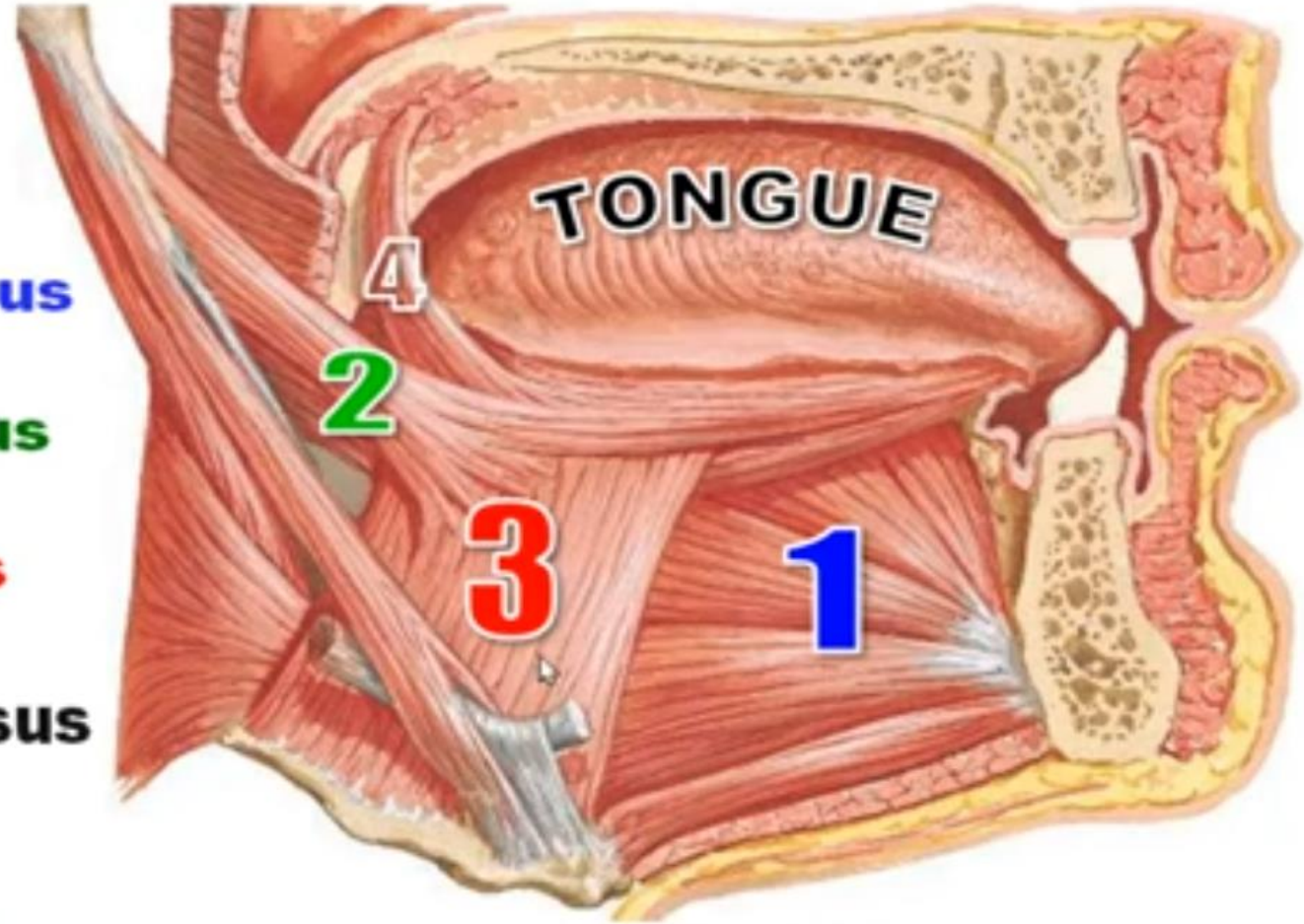
Intrinsic Muscles

1. Longitudinal: superior & inferior
2. Vertical
3. Transverse
4. Oblique



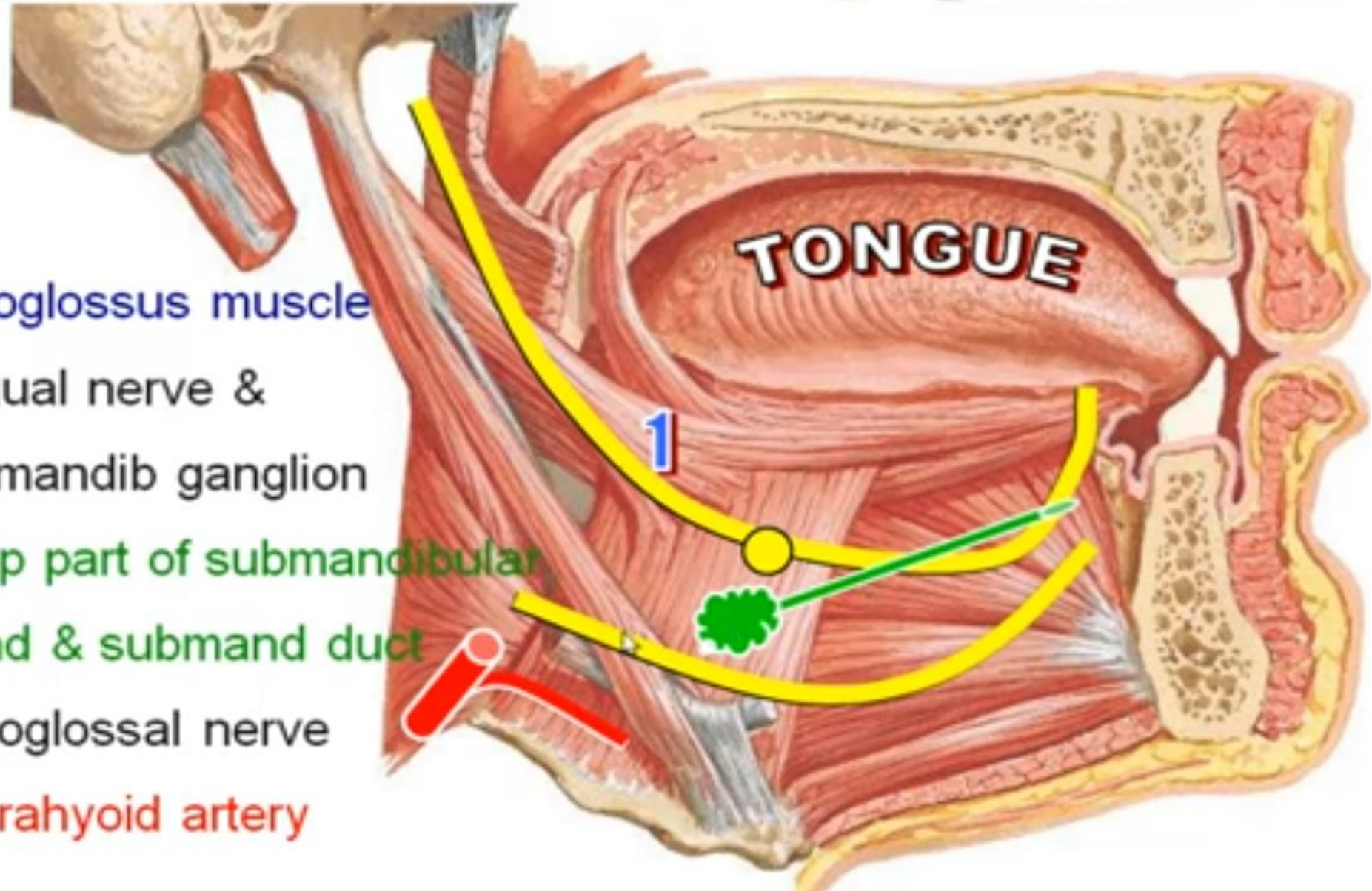
Extrinsic Muscles

1. Genioglossus
2. Styloglossus
3. Hyoglossus
4. Palatoglossus



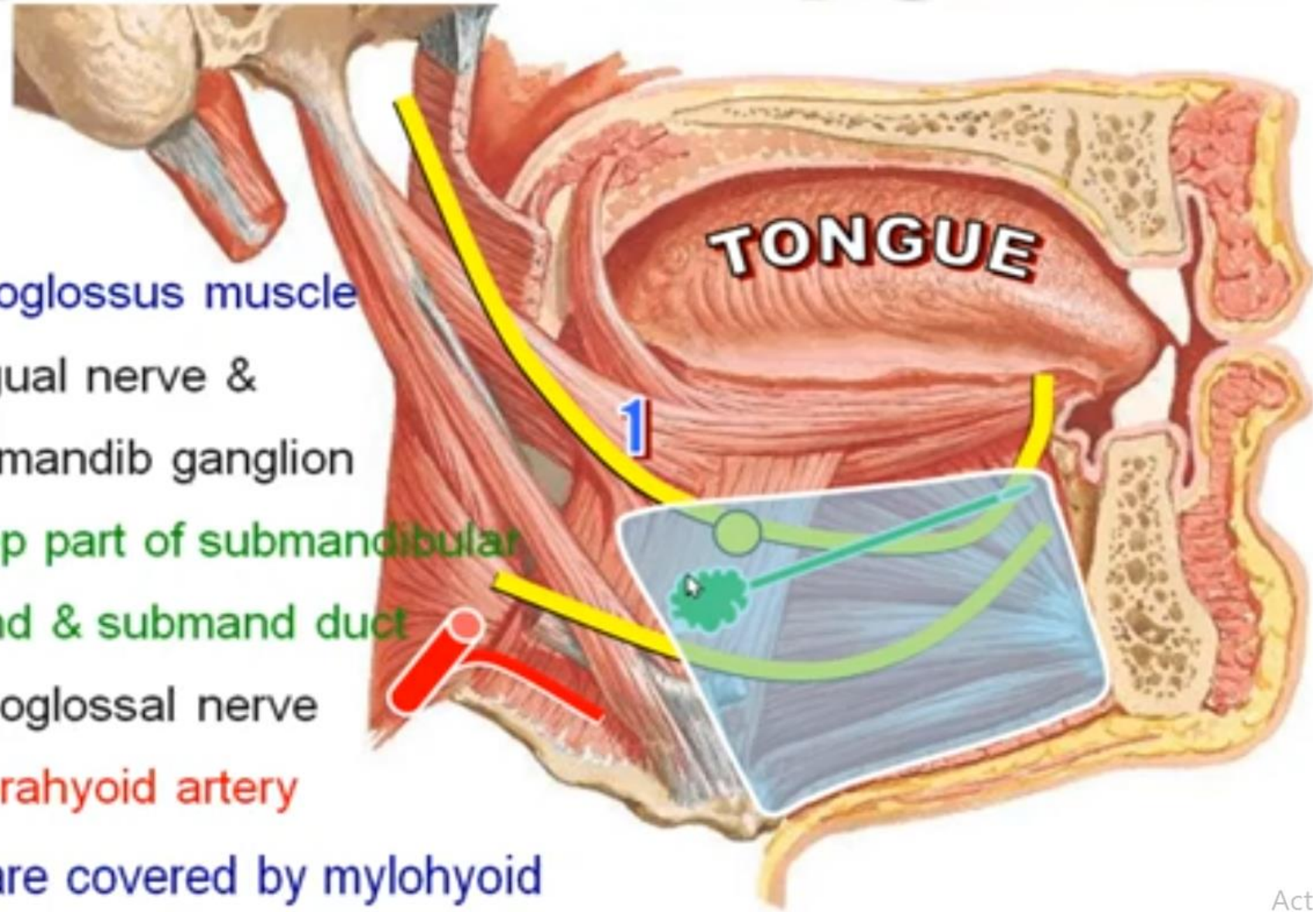
All the muscles are supplied by the XII nerve except palatoglossus which is supplied by the vagus

Superficial Relation of Hyoglossus: 5



1. Styloglossus muscle
2. Lingual nerve & submandib ganglion
3. Deep part of submandibular gland & submand duct
4. Hypoglossal nerve
5. Suprahyoid artery

Superficial Relation of Hyoglossus: 5

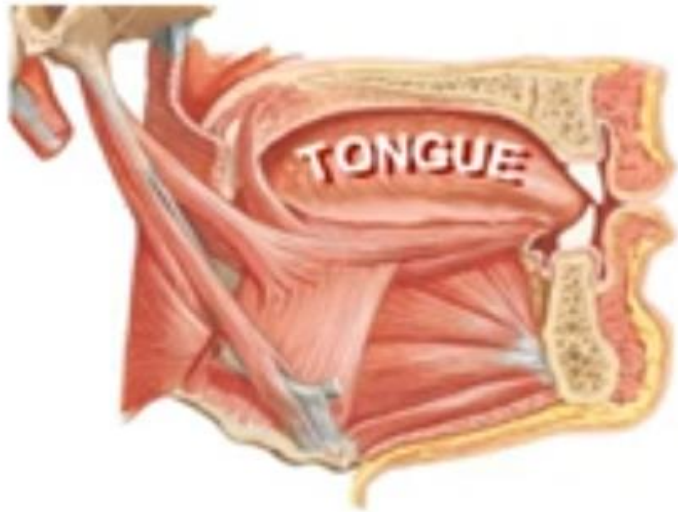


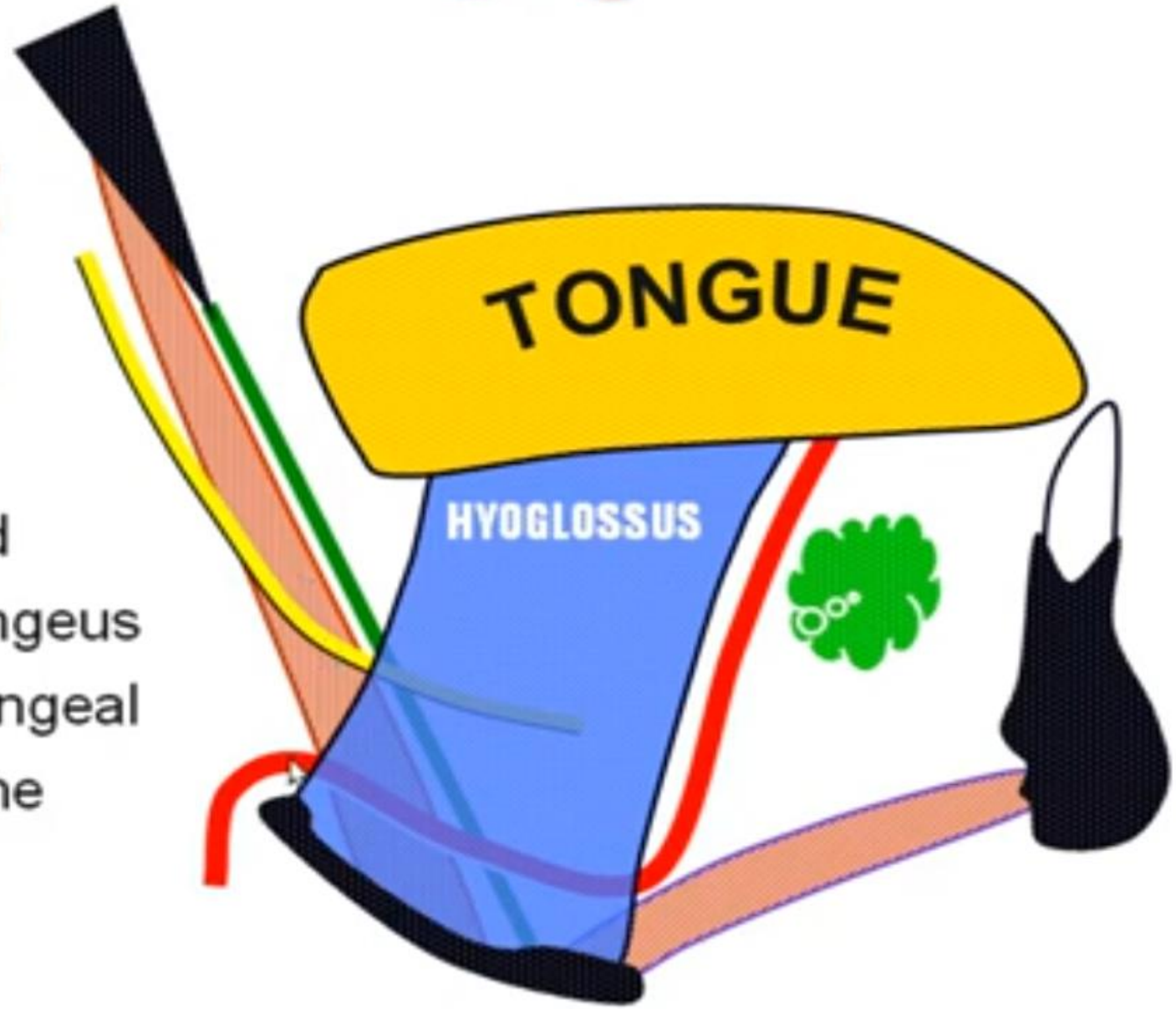
1. Styloglossus muscle
2. Lingual nerve & submandib ganglion
3. Deep part of submandibular gland & submand duct
4. Hypoglossal nerve
5. Suprahyoid artery

They are covered by mylohyoid

Act

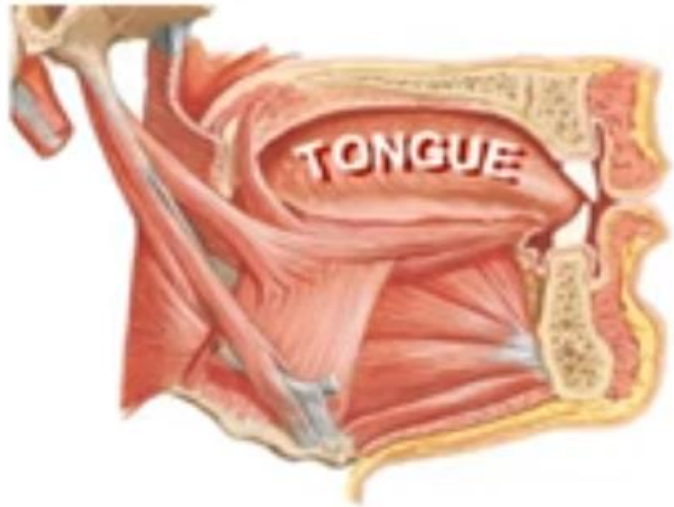
Deep Relation of Hyoglossus: 5



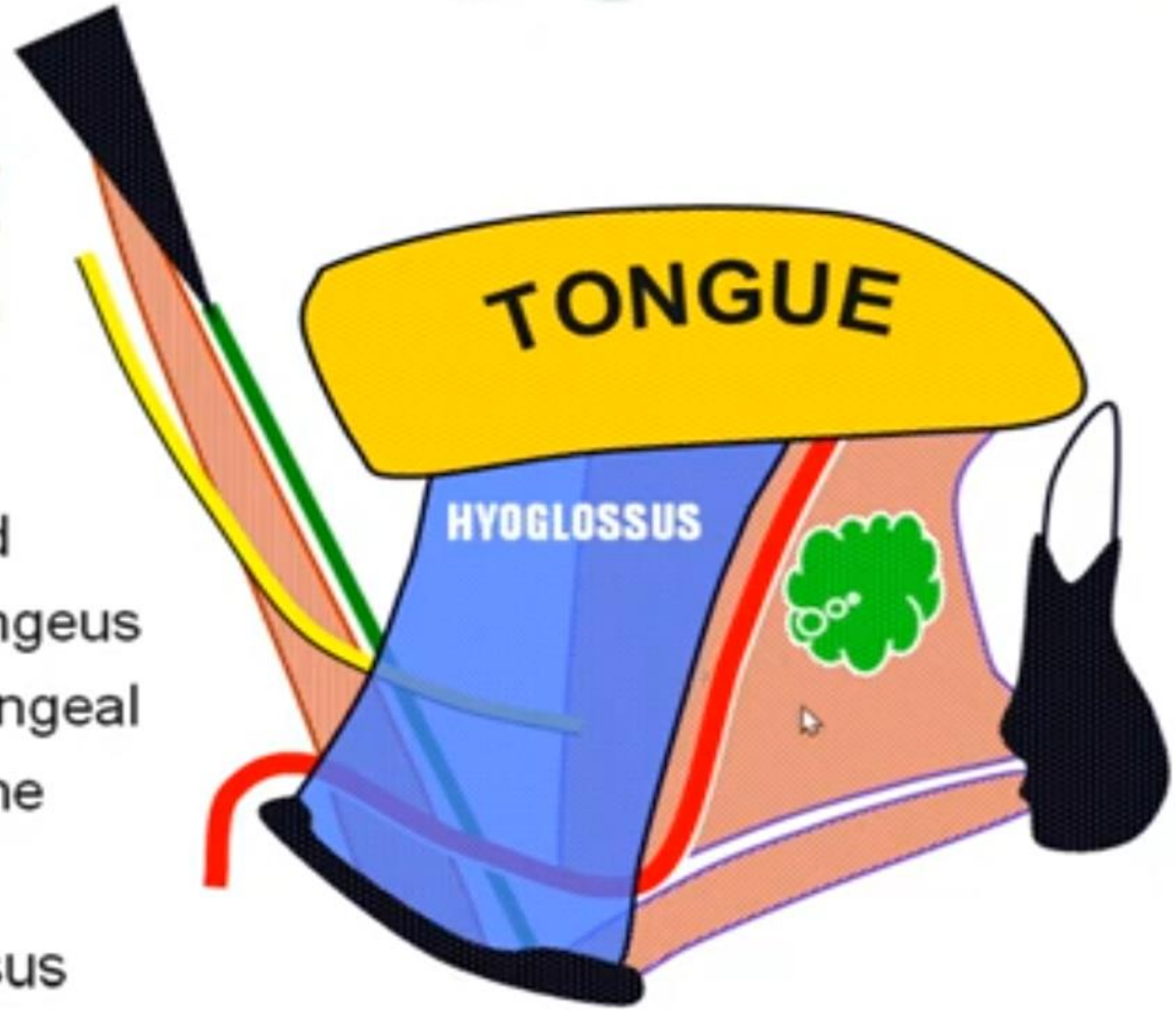


1. **Ligament:** stylohyoid
2. **Muscle:** stylopharyngeus
3. **Nerve:** glossopharyngeal
4. **Artery:** 2nd part of the lingual artery

Deep Relation of Hyoglossus: 5



1. **Ligament:** stylohyoid
2. **Muscle:** stylopharyngeus
3. **Nerve:** glossopharyngeal
4. **Artery:** 2nd part of the lingual artery
5. **Muscle :** genioglossus



NERVE SUPPLY

1. MOTOR:

all the muscles are supplied by the XII nerve except palatoglossus muscle which is supplied by the vagus nerve (pharyngeal plexus)

2. SENSORY:

a. Anterior 2/3:

1. Lingual nerve: general sensation
2. Chorda tympani: taste sensation

b. Posterior 1/3: Glossopharyngeal nerve (general & taste sensation)

c. Most posterior part: vagus nerve



LYMPH DRAINAGE OF THE TONGUE

TIP:

to submental Lymph nodes

ANTERIOR 2/3:

Submandibular lymph nodes

Central part: bilateral

Peripheral part: unilateral

POSTERIOR 1/3:

upper deep cervical lymph nodes

